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CMPE 491 Senior Project

ARIS

High Level Design Report

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# **1. Introduction**

## **1.1 Purpose of the System**

The purpose of ARIS is to provide users a hands-free and AI-assisted system in challenging environmental conditions. System allows users to get and send commands, therefore communicate information through multimodal inputs like facial expressions, hands gestures and voice commands without interacting manually any communication devices. The target users of the proposed system are personnel operating in high risk and safety critical operational environments where hands free interaction and situational awareness are essential. Users may require awareness of the situation as well as physical readiness. This system is generally intended to analyze the user inputs in real time while with guiding them to the central unit or the connected device through artificial intelligence algorithms after processing.

In the analysis phase, the key problem identified is the lack of efficient communication channels that do not distract and pause the motion of the user under high-pressure circumstances. In the design phase, the same problem is overcome by integrating a multimodal AI-based interaction design approach and a modular and scalable system architecture. The proposed design converts the problem definition formulated during the analysis phase into an achievable solution using real-time command recognition, efficient communication, and secure transmission of data while retaining flexibility for future developments in the system.

## **1.2 Design Goals**

The proposed design goals are determined according to the proposed requirements of the system. The following goals meet the purpose of the system and they can be easily applied to real-world applications.

### **1.2.1 Modularity**

One of the important design goals of the system is modularity. The system is divided into separate components including input processing, command-recognition, communication and user interface. This structure allows each module to be developed, tested and improved without affecting the other system components. Also, it simplifies the process of debugging and integration while allowing future extensions.

### **1.2.2 Scalability**

ARIS is designed to have multiple-user operations concurrently, allowing simultaneous data flow between the clients and server. As the number of clients, in other words the connected helmets increases the system should continue to function reliably. This is important because ARIS is designed to be expanded in future work. The use of a centralized server architecture enables the system to scale efficiently while managing multiple active clients.

### **1.2.3 Performance and Latency**

System performance in real-time and low latency is critical for ARIS system. Since the system depends on the input from the user such as hand gestures, facial expressions and voice commands, delays between these and the system response are significant. In time-sensitive

environments even small delays may negatively affect usability and safety. For this purpose, ARIS is designed to minimize latency. The system uses edge processing in this context, allowing inputs to be processed close to their source and converted into commands with minimal delay.

#### **1.2.4 Security and Privacy**

Security and privacy play critical role in the design of ARIS. The system processes audio and visual data that can include personal or operational information which makes data protection an important design goal. To address this ARIS incorporates authentication mechanisms and encrypted communication channels ensuring that information remains secure during data transmission.

#### **1.2.5 Portability**

Portability is a necessary design goal of the ARIS system. Hardware and software components of ARIS are designed to operate independently with a clear separation between them so that ARIS can be deployed on different hardware platforms without requiring major changes to software architecture. As a result system supports software reusability and can be more easily adapted to alternative hardware configurations.

#### **1.2.6 Maintenance and Extensibility**

Maintenance and extensibility are key considerations to ensure long-term usability of ARIS. System is designed to support updates and modifications without introducing unnecessary complexity. This approach allows AI models to be updated, system configurations to be adjusted and new features to be integrated as needed. By having simplicity in design ARIS becomes easier to understand, maintain and extend in future development stages.

### **1.3 Definitions, Acronyms, and Abbreviations**

This section lists the main terms and abbreviations used in the report to avoid confusion and ensure clarity.

- **ARIS:** The AI assisted smart helmet communication system proposed in this project.
- **AI:** Artificial intelligence techniques used for recognition and decision-making process.
- **Input Processing:** The part of where the system collects and prepares data from input sensors such as camera and microphone.
- **Edge Processing:** A method of processing data close to where it is generated to reduce delay.
- **Server:** Central component of the system that is responsible for managing communication between devices.
- **Client:** The helmet unit that sends user inputs and receives responses.
- **Latency:** Time delay between a user action and system response.

- **Encryption:** The procedure used for protecting data from unauthorized access by converting it into another format.
- **Authentication:** The process of confirming the identity of a user or device before allowing access to the system.
- **Command Recognition:** Interpreting user inputs and matching them with predefined commands.
- **User Interface (UI):** Component that displays system status and connection information.

## 1.4 Overview

This document describes ARIS system design at a high-level perspective, including key system design decisions that make up overall system architecture. The next sections elaborate system architecture design, along with system design decision-logic. This document ends with a description of main decisions and future development paths.

## 2. Current Software Architecture

Since ARIS is a newly proposed system, it does not have existing or legacy software architecture.

Current communication solutions used in similar operational environments mainly depend on handheld or manually operated devices, such as conventional radio systems. Although these solutions provide basic communication functionality, they often restrict user mobility and require constant manual interaction, which can be inefficient and potentially unsafe in high-pressure or dynamic situations.

Furthermore, traditional systems lack support for multimodal interaction and real-time AI-based interpretation of user inputs, including gestures, facial expressions, and voice commands.

These limitations highlight the need for a new software architecture that supports hands-free operation, real-time processing and secure communication. Based on these considerations proposed ARIS system architecture is presented in following section.

## 3. Proposed Software Architecture

### 3.1 Architecture Overview

The ARIS system's proposed software architecture is meant to make it possible for people to talk to each other in real time without using their hands in changing operational environments. The architecture is based on a distributed client-server model, which means that the helmet-mounted unit and the central command system share some of the work. This approach enables

low-latency processing at the edge while maintaining centralized control, monitoring, and coordination.

The helmet unit operates as an intelligent edge device responsible for capturing and interpreting multimodal inputs such as facial expressions, hand gestures, and voice commands. To meet real-time performance requirements, input processing and AI-based recognition are performed locally on the helmet. This design reduces communication overhead and ensures that command recognition remains responsive even under unstable network conditions.

The central server acts as the backend coordination layer of the system. It receives validated commands from multiple helmet units, manages secure communication sessions, and provides real-time monitoring through a web-based dashboard. System logs, telemetry data, and feedback messages are also handled at this level, allowing command center operators to maintain situational awareness and system oversight.

Overall, the ARIS architecture emphasizes modularity, scalability, and security. Each major function of the system is clearly separated, enabling independent development and future extensions without affecting the overall system structure. This high-level design ensures that ARIS remains adaptable to evolving requirements while preserving reliable and efficient operation.

## **3.2 Subsystem Decomposition**

This will be accomplished by the following subsystems identified to form the ARIS system, with each subsystem managing a particular aspect of the system. This will ensure modularity, ease of maintenance, and a clear division of labor in the overall structure.

### **3.2.1 Input Processing Subsystem**

It is necessary that all the video and audio information coming from the helmet unit is processed by the input processing subsystem. The input processing subsystem relies on a Raspberry Pi 4 Model B, a system that maintains performance, power, and availability. It uses the Raspberry Pi Camera Module v3, which captures video. It uses the default microphone, which captures the audio.

The primary application on the helmet device is primarily concerned with handling the captured data from sensors. This application manages data acquisition, buffering, as well as initial processing of audio inputs and camera images. The network-based interfaces make it possible for any communication to occur between components responsible for inputs and the overall system so that data flow is possible.

The collected information is made to undergo basic preprocessing techniques before any other processing. These techniques, to ensure the quality of the information under different environmental conditions, include frame preparation for visual signals as well as noise removal for audio signals. After the helmet has been switched on, the subsystem runs in the

background without the need for any human interaction. The system design ensures uninterrupted information flow without any effect on latency.

### **3.2.2 Command Recognition Subsystem**

The Command Recognition Subsystem is tasked with analyzing preprocessed visual and audio inputs using artificial intelligence-based command recognition models. This subsystem recognizes hand gesture commands, face commands, and pre-defined voice commands separately and then processes the outcomes to deduce commands from the user.

The recognition procedures are carried out within the helmet unit to make the responses instant and independent of the constant availability of the network. In the process of each recognized input, there is a corresponding confidence level, which is sent to the validation stage.

### **3.2.3 Command Validation and Mapping Subsystem**

This is another subsystem that tests the structural validity and confidence levels of recognition outputs. Its primary task is to ensure that only those recognition outputs that contain trustworthy and well-defined commands should be allowed to further proceed in the system.

The recognized inputs are then linked to predefined definitions in the command dictionary after being validated. The process of linking prevents ambiguous or unintended actions and ensures consistency in the interpretation of commands. The inputs with levels of confidence lower than the required threshold are either discarded or tagged for repetition without being sent to the engine or service.

### **3.2.4 Communication Subsystem**

The communication between helmet unit and central server is secure owing to the Communication Subsystem. This subsystem employs encrypted network connections to handle the transmission of feedback messages, system telemetry, as well as verified commands.

The subsystem allows for temporary offline usage through buffering of generated commands in case of lost connectivity, ensuring data integrity even where network connectivity is not stable. The data in the buffer gets synchronized once connectivity is restored.

### **3.2.5 Backend and Monitoring Subsystem**

This is because the Backend/Monitoring Subsystem is responsible for managing roles in the system and revolves around operations that take place on the server side of the system. It is responsible for handling commands as well as other information coming from multiple helmet units.

The services provided by this subsystem through backend services allow access to historical and real-time system updates of system status. The ability to centrally monitor and control helps the operators evaluate the system's effectiveness and provides them with situational awareness of the system's status.

### 3.2.6 User Interface Subsystem

The User Interface Subsystem shall present information that is related to system and provide the user through a web-based dashboard with information regarding the current status. Presenting this information will be in an orderly and understandable manner, regarding commands, alerts, and statuses in real-time.

The interface allows users to monitor linked helmet units, study logs, and observe system behavior without touching the helmet devices directly. Its design gives a very high priority to clarity and rapid access to information, so that effective decisions can be made with the operational use of the system.

## 3.3 Hardware / Software Mapping

The ARIS project features the architecture of a distributed client-server, which distributes computational load and responsibilities between Edge Node and Server Node. The mapping from hardware to software components is given by the Deployment Diagram, structured as follows:

1. ARIS Helmet Unit (Client Node - Edge Computing): The smart helmet, as the field unit, uses a Raspberry Pi 4 embedded system designed to provide portability and also satisfy economic constraints.

- Hardware Components: Raspberry Pi - Processing Unit; Camera Module - Visual Input; Microphone Array - Capture Audio Commands
- Hosted Software Artifacts:
  - InputManager: It continuously listens to raw streams coming from the sensors.
  - VisionProcessor and AudioProcessor: These two functions perform tasks that involve image processing algorithms as well as audio analysis algorithm tasks. To minimize the time to less than 2 seconds, the AI inference is performed on the device.
  - CommunicationModule: It encrypts processed and authenticated commands, which then are relayed server.

2. Command Center Unit (Server Node): This can be defined as the central unit where operational monitoring occurs.

- Hardware Components: High capacity central server
- Hosted Software Artifacts:
  - WebInterfaceController: Enables the Dashboard user interface used by commanders.
  - LogManager & FeedbackHandler: It is responsible for the write operations of the incoming data into the database as well as the feedback (ACK) signals back to the helmet.

Communication Channel: The connection between the two points takes place through an encrypted wireless channel (TCP/IP over Wi-Fi/Radio) that uses AES-256 standards.

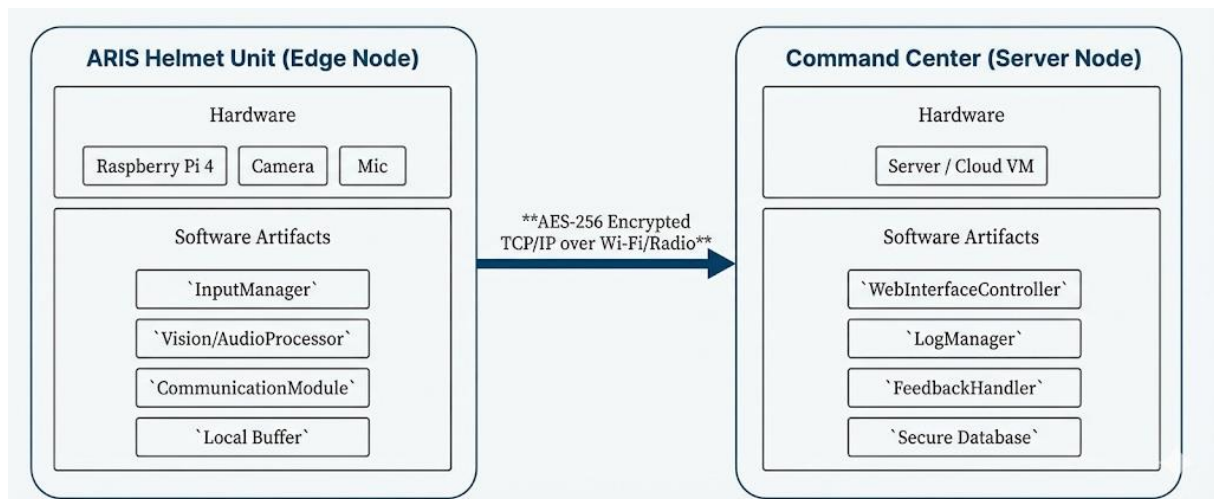


Figure 1 – ARIS Architecture

### 3.4 Persistent Data Management

The system of ARIS follows a “Selective Data Retention” approach and adheres to military security requirements and principles of ethical privacy. Data Management is classified into Transient, Persistent types:

#### 1. Transient Data (Not Stored):

- **Raw Biometric Data:** Real-time video frames and audio streams captured by the camera and microphone are never permanently stored on the disk. These data are loaded into The InputManager reads input from hardware into memory; the input is then translated into command signals by the RecognitionEngine, and immediately discarded once the processing is complete.
- **Justification:** According to privacy constraints outlined in Analysis Report, storing soldier facial and voice recordings sets an ethical risk. Technically, storing continuous video on a Raspberry Pi would deplete storage rapidly and degrade system performance, hindering the real-time processing goal.

#### 2. Persistent Data (Stored):

- **System Logs.** The information including command type, timestamps, transmission status, and system errors is stored in the secure database on the central server through the LogManager.
- **Configuration Files:** The system’s calibration parameters, including audio sensitivity or gesture thresholds, along with the Command Dictionary, will be stored in the local file system through the use of ConfigManager. So, the system will not need to be calibrated every time it is rebooted.
- **Data Management:** Data is organized using a relational database or encrypted JSON/XML files.

While raw biometric data such as video and audio streams are not stored, the system retains anonymized event level metadata (e.g., event type, timestamps and confidence scores) for

system monitoring and safety analysis. This data does not allow the reconstruction of biometric information and complies with privacy by design principles

### 3.5 Access Control and Security

In order to guarantee safe working, the correctness of the data and the ARIS system's access grant control also multiple access control and security method is employed. Because of the real time command transmission and monitoring's sensitive nature, the system design encompasses very strong authentication, authorization and encryption mechanisms.

#### Access Control Mechanisms

**Role Based Access Control (RBAC)** is implemented to restrict system functionality based on user roles:

- **Soldiers (Helmet Users):**  
Only able for sending command, receiving feedback and monitor their own helmet status. Soldiers are not allowed to see the system logs or access other users' data.
- **Command Center Operators:**  
Permitted to observe the current helmet status, incoming commands, alerts and operational logs in real time. They can evaluate field activity but cannot alter system level configurations.
- **Root Users (Administrators):**  
Possess complete power over system parameter, user roles, recognition models and command lexicons. Administrator only functions are limited to identity verified root users.

#### Authentication

All users accessing the ARIS web interface are authenticated using secure login mechanisms. In order to avoid any unauthorized system access, multi factor authentication (MFA) is utilized for administrative and command center access. Helmet devices authenticate themselves to the central server so that reliable communication can be set up at the start of a session.

#### Security Protocols

1. **Data Encryption:**  
All data transmitted between ARIS helmets and the central command system is protected using end to end encryption. Secure communication channels (e.g., TLS 1.3) are used to protect commands, telemetry data and also feedback messages during transmission.
2. **Secure Communication Interfaces:**  
All the APIs that are server side use HTTPS only to make sure that data cannot be intercepted, tampered with or replayed. The use of encrypted sessions provides confidentiality and integrity of the operational data.

- 3. Auditing and Logging:**  
System activities like command transmissions, authentication events, errors and alerts are logged securely. The logs serve for traceability, post operation analysis and security audits but are still restricted to authorized users only.

## **Compliance and Privacy**

The ARIS system is designed in compliance with data protection regulations such as GDPR. Collected visual, audio and also operational data are processed strictly for system functionality. They are not stored beyond operational requirements. Anonymization techniques are applied to logs and monitoring data where applicable to protect user privacy.

## **3.6 Global Software Control**

In order to achieve real-time reactivity and energy efficiency, the control flow process for the ARIS system is realized through the principles of the 'Event-Driven Architecture'. The system does not follow a procedural workflow logic, as it relies on a 'Processing Pipeline' triggered by external events.

Control Logic and Flow:

1. Event Detection: In the Idle state, the InputManager is constantly observing the environment. Upon detecting a certain movement or sound event (Threshold Trigger), the system moves to the "Active Processing" state.
2. Decision Making: The control shifts to the RecognitionEngine. The input is processed, and based on the analysis, the Confidence Score is determined. If the Confidence Score surpasses the "90% threshold," then the CommandValidator triggers the "Command Transmission Event."
3. Asynchronous Communication: The communication component of the system performs asynchronous communication with the server so that the system does not have to wait while communicating with server.
4. Error & Exception Handling (Boundary Conditions): In the event that the wireless link is lost, the global control state switches to the Offline Mode. Rather than transmitting, data is stored in the local buffer. Then, following the occurrence of the 'Connection Restored' event, automatic synchronization begins.

Summary: The software control is not based on one main loop but is a dynamic structure driven by "Input Events" from sensors and "State Events" (ACK/NACK) from the network.

### **3.7 Boundary Conditions**

This section defines the operational boundaries, startup and shutdown conditions and also error handling mechanisms of the ARIS system. These conditions ensure stable operation of the helmet based system under varying environmental, network and hardware constraints.

#### **3.7.1 System Initialization**

During system startup ARIS performs the following operations:

- Initialize helmet hardware components including camera, microphone, communication module and onboard processing unit.
- Preloads the AI recognition models as well as their respective config parameters into the memory.
- Verifies the protected connection between the helmet and the control server.
- Sets up encrypted channels for the traffic of telemetry and command signals.
- Performs mandatory hardware diagnostics (battery condition, temperature, sensor presence).
- Switches the system to operational mode after all tests are successfully done.

If any critical component fail during initialization the system prevents full activation and notifies the user accordingly.

#### **3.7.2 Normal Operation**

Under normal operating conditions ARIS functions as follows:

- Continuously captures and processes visual and audio inputs in real time.
- Executes AI based recognition for gestures, facial expressions and also for voice commands.
- Secure transmission of validated documents and telemetric data takes place from the command center to the unit.
- The unit gets feedback or directions from the central part and then the user receives them.
- The overall performance of the system is tracked through latency, recognition confidence, signal strength and also temperature metrics.

The noise, movement and also lighting conditions that are moderate will not stop the system from operating continuously as it is designed to do so.

#### **3.7.3 Boundary and Operational Limits**

The following boundary conditions define safe and reliable operation of the ARIS system:

- Minimum and maximum camera resolution limits are enforced to maintain recognition accuracy and real time performance.

- Environmental constraints such as lighting conditions, dust levels and visibility directly affect recognition accuracy . They are considered during operation.
- The maximum operating temperature and battery threshold is set to avoid hardware damage or overheating.
- The network latency and bandwidth limit are monitored continuously to ensure command delivery within the predefined response time.
- Recognition confidence level are strictly maintained to prevent the transmission of wrong or unclear commands.
- Local buffering of commands provides temporary offline operation support during connectivity loss.

### 3.7.4 Error Handling and Recovery

ARIS incorporate multiple error detection and recovery mechanisms:

- **Authentication Errors:**

When user authentication is not successful the system will show a clear error message and restrict access.

- **Communication Failures:**

If there is an interruption in the network or the server is down, the system will go to offline mode and save events on the local drive and try to connect again automatically.

- **Sensor or Hardware Failures:**

The malfunctions in the camera, microphone or power components are detected through the continuous monitoring process. The user and command center are informed straight away.

- **AI Processing Errors:**  
If recognition confidence falls below acceptable thresholds, the system requests input repetition instead of transmitting unreliable commands.

All error are logged with timestamps to support diagnostics and post operation analysis.

### 3.7.5 Shutdown Sequence

ARIS ensures that upon system shutdown:

- All the active communication sessions are properly ended in a secure manner.
- All the pending command and telemetry data are either saved or synchronized if the connectivity permits.
- The system logs and configuration states are securely kept.

- The hardware parts are turned off in safe manner so that data corruption or physical damage will not occur.

The shutdown procedure is intended to be both polite and resistant to faults. Thus it will maintain system integrity even if the power goes off unexpectedly.

## 4. Subsystem Services

### 4.1 Multimodal Command Recognition Service

The Multimodal Command Recognition Service handle the detection and interpretation of user commands that are based on audio and visual inputs being captured by the helmet. The data from the camera and microphone are first preprocessed and then later sent to the AI Recognition Engine. It detects the gestures, facial expressions or voice commands that have been predefined and gives them confidence scores.

The confirmed commands go through process of validation for reliability before they are sent in a secure manner to the Command Center via the Communication Module. At the same time system telemetry and status information are being sent to the Web Dashboard so that monitoring in real time and operational awareness are possible.

Figure 2 presents the high level architecture of the Multimodal Command Recognition Service and illustrates the interaction between input components, AI processing modules, validation logic and also communication interfaces.

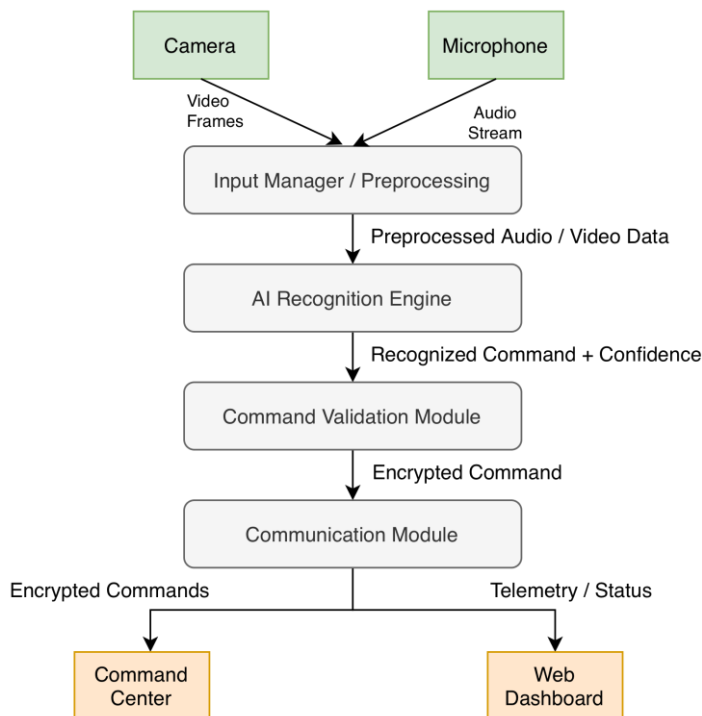


Figure 2 – Multimodal Command Recognition Service Architecture

## 4.2 Proximity Detection Service

Although originally motivated by military use cases, these safety related services are designed to be generic and applicable to various high risk operational domains, including industrial sites, emergency response scenarios and field operations.

The Proximity Detection Service is aimed at improving employee safety in the workplace by tracking the proximity between workers and dangerous machines. The system uses geometric calculations and camera calibration technique to determine the distances in the real world based on the bounding box data generated by the object detection module. The computed distances are constantly checked against the safety limits set beforehand.

The system notifies users instantaneously when a risky proximity is detected. The notifications are sent to the backend through REST APIs for the purpose of logging and analysis. They are also sent to the frontend dashboard through WebSocket connections for visualization and user notification right away.

Figure 3 illustrates the high level workflow of the Proximity Detection Service, including distance calculation, risk analysis and alert generation components.

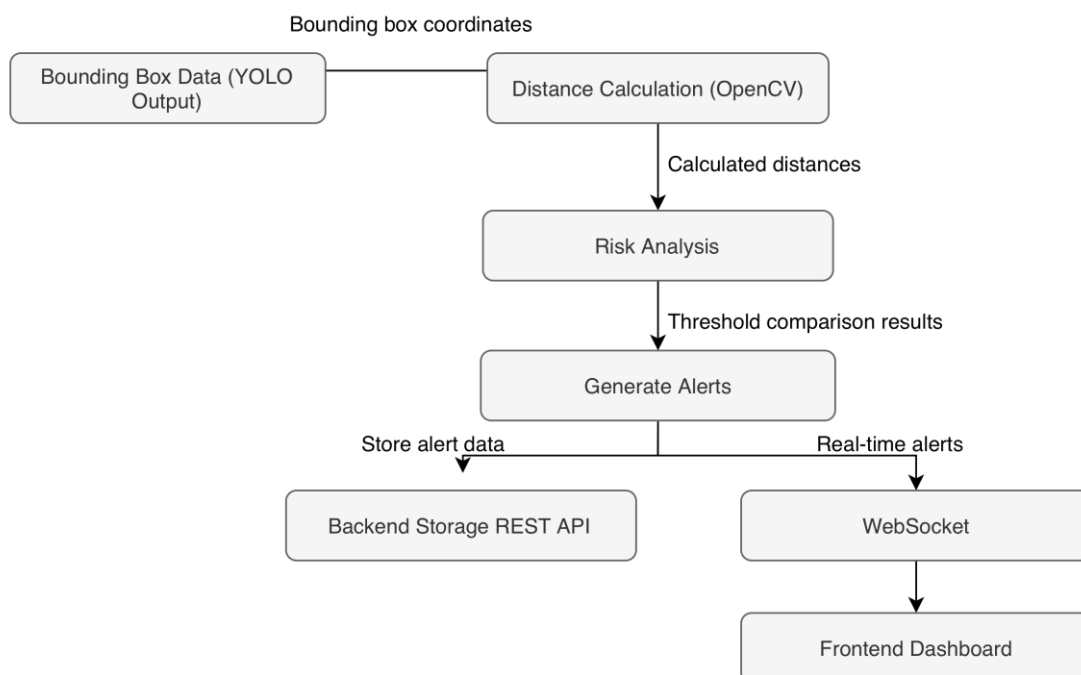


Figure 3 – Proximity Detection Service Architecture

### 4.3 Backend Service

The Backend Service serves as the main coordinating layer of the ARIS system, managing data storage, secure communication and operational processing. It gets authenticated commands and telemetry data from the services on the helmet side. It keeps the command center and the web dashboard up to date with real time data by logging critical events and system logs and providing access to them. REST APIs are provided for configuration management, historical data retrieval and user role operations while WebSocket is employed for real time status and alert streaming. Sensitive information is secured with encryption during transmission and access controlled methods and thus guaranteeing a dependable and synchronized operation among all ARIS components.

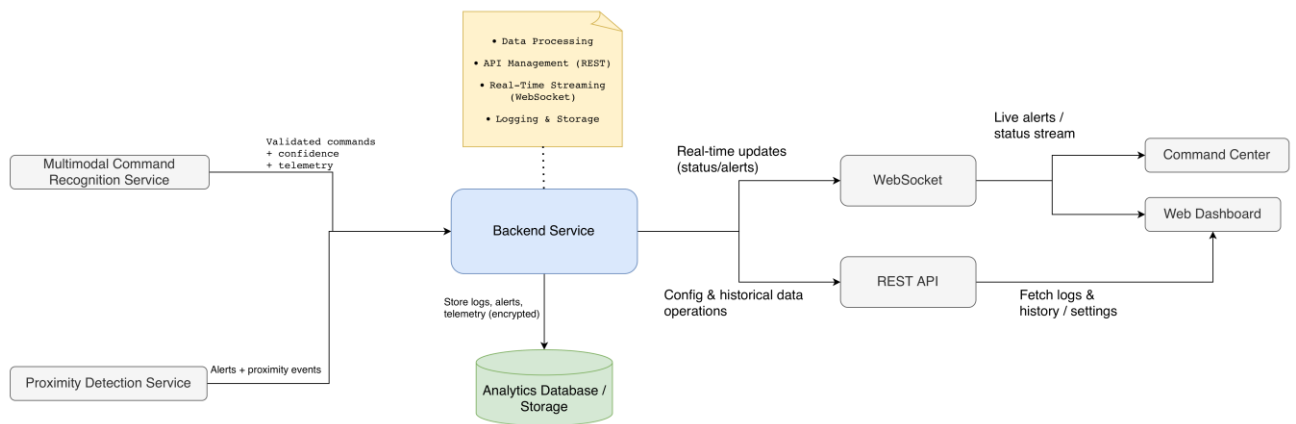


Figure 4 – Backend Service Architecture

### 4.4 Real Time Communication Service

The Real Time Communication Service is one of the main actors that delivers notifications in real time. For instance updating system statuses and also controlling the flow of data from the backend to the user interfaces. For safety critical events such as proximity alerts and command execution feedback it employs WebSocket to ensure very low latency communication. The service uses REST API for non real time operations such as getting logs, changing system settings and accessing historical data. This division of operations guarantees communication that is competent, dependable, and scalable throughout the system.

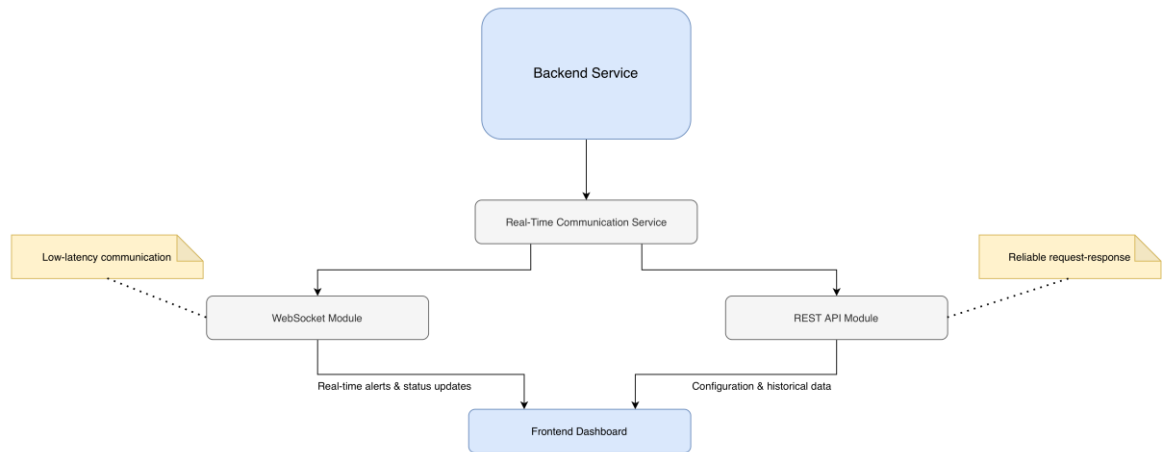


Figure 5 – Real-Time Communication Service Architecture

## 4.5 Safety Analytics & Compliance Tracking Service

The Safety Analytics and Compliance Tracking Service is extremely important in the examination of past safety data and the granting of precious insights about the compliance to safety regulations in the workplace over the years. It also takes in and also synthesizes the data produced by the PPE Detection Service, Proximity Detection Service and System alert logs to give holistic view of the safety performance. The service processes this data in such a way that it is able to point out the places where strict safety rules are every now . Then broken, periods that risk the most and areas of the work environment that are most affected frequently.

This service gives the safety officer and administrator the support they need by supplying them with summarized compliance metrics, trend analysis and severity based reporting. Historical, anonymized safety related event data is retained for long term trend analysis and compliance evaluation. It can be accessed for comparison and evaluation of performance over a long time. The analyzed outcomes are sent to the frontend dashboard where users can see the compliance rates. Later they look at the past incidents and check whether there are any improvements in or deteriorations of safety behavior. By changing the raw detection data into structured analytics, this service has made crucial contribution to decision making based on data and the ongoing improvement of workplace safety.

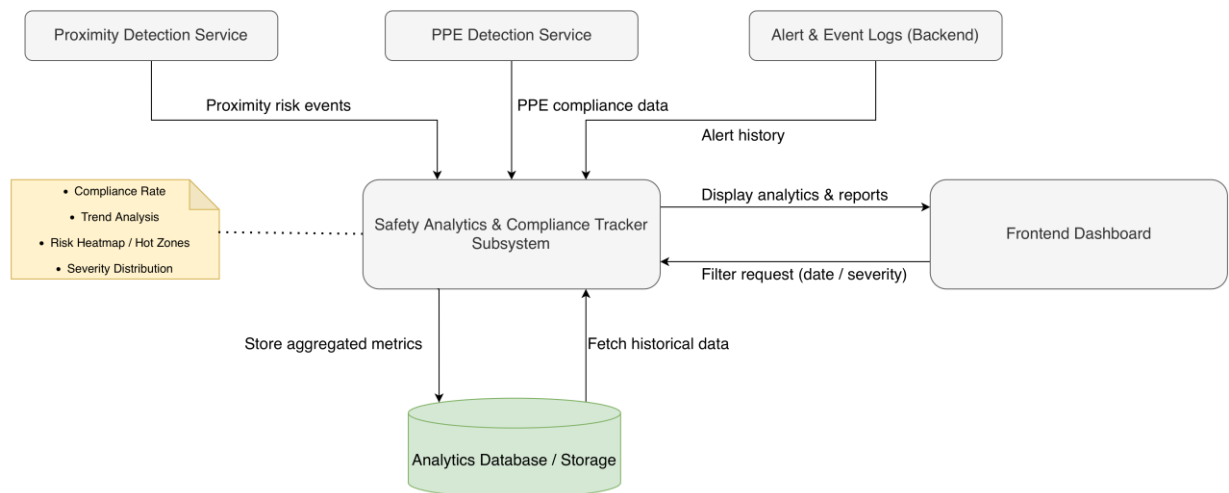


Figure 6 – Safety Analytics and Compliance Tracking Service Data Flow Diagram

## 5. Glossary

- **AI (Artificial Intelligence):** The ability of system to deal with such human like activities as detection, analysis and also even making decisions.
- **Authentication:** The act of confirming the user's or system's identity prior to allowing the resources.
- **Backend Service:** The technological infrastructure located at the server side that is responsible for data processing, alert generation, communication management and logging for security purposes.
- **Data Visualization:** Graphical representation of safety associated data, alerts and analytics to aid monitoring and decision making.
- **Encryption:** A method of data protection which involves changing the information into code that only the authorized people can decipher.
- **Frontend Dashboard:** The graphical interface through which real time alerts, system condition, analysis and reports on occupational health and safety are presented to the user.
- **Machine Learning (ML):** A branch of artificial intelligence that facilitate the development of models capable of detecting objects, PPE compliance and safety related patterns from visual data.
- **Object Detection:** A computer vision algorithm focus on the identification and localization of visualized objects like people and protective gear in photos or video clips.
- **PPE Detection:** The analysis of whether employees have donned the recommended safety equipment by means of visual computer methodologies.
- **Proximity Detection:** The system that determines insecure distances among workers, equipment or hazardous area.

- **Real-Time Communication:** Nonstop flow of information between server side applications and client side interface without any perceptible lag.
- **RBAC (Role-Based Access Control):** A technique for restricting access to a computer system based on the roles and rights of users that is utilized as a security safeguard.
- **Scalability:** The property of the system whereby it can cope with the rising amount of data, users or event without compromising on speed and quality.
- **WebSocket:** It is a communication protocol that allows continuous and bidirectional transfer of data for real time notifications and updates.
- **YOLO (You Only Look Once):** An object detection algorithm that works in real time. It is fast as well as accurate in the case of the video streams.

## 6. References

- **Redmon, J., Divvala, S., Girshick, R., & Farhadi, A. (2016).**  
*You Only Look Once: Unified, Real-Time Object Detection.*  
Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR).
- **OpenCV Team. (2024).**  
*OpenCV Documentation.*  
<https://docs.opencv.org>
- **Mozilla Developer Network (MDN). (2024).**  
*WebSocket API Documentation.*  
[https://developer.mozilla.org/en-US/docs/Web/API/WebSockets\\_API](https://developer.mozilla.org/en-US/docs/Web/API/WebSockets_API)